



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Computer Science and Business Systems

Academic Year 2024 – 2025 (Even Semester)

Degree, Semester & Branch : IV Semester B.Tech CSBS
Course Code & Title : CS3492 Database Management Systems
Name of the Faculty member : Dr.S.Erana Veerappa Dinesh, ASP / CSBS

Innovative Practice Description

- **Unit / Topic:** Unit II / Dependency Preservation -Boyce/ Codd Normal Form
- **Course Outcome:** CO2
- **Activity Chosen:** Think Pair Share

- **Justification:**

The understanding of database normalization, particularly the Boyce-Codd Normal Form (BCNF), is essential for ensuring minimal redundancy and avoiding anomalies in database design. BCNF helps students identify functional dependencies and refine the structure of relational schemas to achieve a higher level of consistency and efficiency. By working through normalization, students develop a logical approach to organizing data, ensuring data integrity, and optimizing database performance. This activity encourages critical thinking and collaboration as students analyze and validate their design choices with peers, reinforcing their understanding of proper database structuring.

- **Time Allotted for the Activity:** 30 Minutes

- **Details of the Implementation:**

A Collaborative, active learning strategy, in which students work on a question posed by instructor, first individually (Think), Write in a paper, then work in pairs or groups(Pair) to solve the problem, and finally share their solution with the entire class (Share).

- **Think**–Write-Students were provided with questions on BCNF. The student has to think individually and write the answer in a paper and it is shown in the Fig 1.

- **Pair**-After writing the answers, the students were allowed to discuss with neighbor and justify the resultant answer to each other. The pair with neighbor was shown in Fig.2

- **Share**-Once the conclusion is made between the pair, then it is to be shared among the whole class by any of the student from the team and it is shown in the Fig 3.

- **CO–PO/PSO mapping:**

CO	PO1	PO2	PO3	PO9	P10	PSO1
CO2	3	3	2	2	3	3

(1– Low

2 – Moderate

3 – High)

- **PO/ PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO9	PSO1
	3	3	2	2	1
Justification for correlation	The students apply the knowledge of dependency preservation in BCNF	The students analyze the given problem statements.	The students evaluate given problem statements and identify solutions using BCNF.	The students demonstrate problem-solving skills as a team.	The students use the BCNF rules that can apply to create an inconsistent database for any real-world business problems.

- **Images/screenshots of the practice:**



Figure 1 Think the solution



Figure 2 Sample Pair



Figure 3 Nithikasree K Shared among II CSBS

Reflective Critique:

❖ Feedback of practice from students and other stakeholders:

- Some of the students write answers to the questions easily and share their ideas with others.
- Some of them need more explanation and examples to get clarity on the BCNF.

❖ Benefit of the practice:(E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- It helps the students to share ideas with classmates and builds oral communication skills. They completed the task by coordinating with each other and got more confidence and a clear understanding in this topic when discussing it with their peers.

❖ Challenges faced in implementation:

- The students who didn't understand the concept is not able to complete it within the time.
- Involve all the students to be active in this activity is a challenging.

References:

- ❖ https://www.ritrjpm.ac.in/images/CSBS/2022-2023/TPS_AD3251_DS_KU_23_24.pdf
- ❖ https://www.ritrjpm.ac.in/images/CSBS/2022-2023/SEV_GE3151_ThinkPairShare.pdf
- ❖ https://www.ritrjpm.ac.in/images/CSBS/2023-2024/TPS_CCS339_CBT_BY_23_24.pdf